# Multivariate Socioeconomic Stress Modeling and Poverty-Risk Forecasting in Armenia

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## Problem Statement

Armenia exhibits persistent regional socioeconomic disparities that manifest through high poverty incidence, fluctuating crime rates, and uneven health-system capacity across marzes. These factors interact to form what the academic literature describes as multidimensional socioeconomic stress which is an emergent condition driven by economic vulnerability, public safety instability, and healthcare resource limitations. Reports from the National Statistical Committee of Armenia indicate significant variations in poverty levels between marzes, with northern and border regions consistently experiencing higher deprivation levels. At the same time, health-system capacity, measured through medical staff density and hospital bed availability, remains disproportionately low outside Yerevan, increasing local vulnerability to health shocks. Crime patterns likewise show heterogeneity across regions, often correlating with economic instability and limited access to social infrastructure.

Despite the availability of these indicators, there is no unified analytical framework that integrates crime, poverty, and medical system capacity into a single measurable construct, nor any predictive system capable of forecasting next-year poverty risk using multivariate lagged indicators. Current government reporting remains isolated by sector, limiting the ability to detect cumulative stress or emerging poverty risk zones. Academic research consistently highlights the importance of composite indices and explainable modeling for evidence-based governance; however, Armenia lacks such a tool. Thus, this capstone aims to develop a scientifically grounded Stress Index and a transparent poverty-risk prediction model to support informed regional decision-making.

**Keywords:** Socioeconomic Stress, Poverty Forecasting, Crime Analytics, Healthcare Capacity

# Project Overview

This project proposes the development of an integrated analytical system that consolidates poverty, crime, and health-capacity data into a clean marz-level panel for the years 2022–2024. The constructed Stress Index will quantify total regional strain by standardizing and combining poverty incidence, crime rates per 100,000 population, medical staff density per 10,000 residents, and hospitalization or bed availability metrics. The index will serve as a diagnostic tool that highlights regional disparities and identifies where systemic vulnerabilities are persistent or intensifying.

In parallel, the project will implement predictive modeling techniques to estimate next-year poverty risk using lagged socioeconomic indicators. These models will be augmented with SHAP (SHapley Additive exPlanations) explainability, ensuring transparency regarding the contribution of each variable to predicted risk. An interactive Streamlit dashboard will present maps, trend analyses, model outputs, and feature attributions, enabling policymakers, researchers, and development organizations to visually explore stress trajectories and structural differences between marzes. The tool is designed with academic rigor and practical usability to address real-world policy needs.

# Minimum Viable Product (MVP) Features

The system will begin with automated extraction and cleaning of poverty, crime, and health-capacity datasets from ArmStatBank. All indicators will be normalized per capita to eliminate population-size distortions. The Stress Index will be constructed using standardized z-scores, allowing direct comparability across metrics and regions. A sensitivity analysis will evaluate whether PCA-based weighting provides additional explanatory power over equal weighting.

The predictive modeling component will rely on lagged variables such as previous-year crime rate, medical staff density, bed capacity, and hospitalization rates. The model will forecast next-year poverty incidence or assign a binary high-risk classification. Out-of-time validation will be applied to ensure robustness. The Streamlit dashboard will present an interactive choropleth map, a year slider for temporal exploration, a marz-level trend panel, and SHAP-based explanatory visualizations to highlight variable importance.

# Technology Stack

The technological foundation of the project emphasizes reproducibility, scalability, and academic transparency. Python will serve as the primary language for data processing, statistical modeling, and machine learning, utilizing libraries such as Pandas, NumPy, SciPy, and Scikit-learn. SHAP will be implemented for model interpretability. PCA, clustering algorithms, and regression frameworks will form the analytical core of the modeling process.

Visualization and public-facing interaction will be achieved through Streamlit, which will host the analytical interface and dynamic maps using Armenia's GeoJSON boundary files. The system may be containerized using Docker for deployment consistency. All code and processing scripts will be version-controlled to ensure full reproducibility and academic auditability.

# Expected Outcomes

The project will deliver a unified socioeconomic dataset that merges poverty, crime, and health-system capacity indicators into a clean marz-year panel. The Stress Index will allow systematic comparison of regional strain, revealing chronic stress zones and temporal patterns. Cluster analysis will generate structural profiles of high-stress and low-stress marzes. The poverty-risk prediction model will provide early-warning insights into potential increases in poverty levels, supported by transparent SHAP explanations that clarify which variables drive risk.

This integrated framework will significantly improve the analytical capacity of governmental and academic institutions by shifting from fragmented indicator tracking to multivariate modeling. The dashboard will serve as both an educational and decision-support instrument, enabling rapid identification of vulnerabilities and supporting targeted policy intervention.

# Practical Importance

This project occupies a critical intersection between economic analysis, public health metrics, and data science. By unifying disparate indicators into a coherent Stress Index, the system captures the cumulative effects of poverty, health-system fragility, and crime on regional well-being. The predictive component introduces a forward-looking dimension that supports strategic planning rather than reactive policymaking. SHAP interpretability ensures that the modeling process remains transparent, allowing policymakers to understand causative pathways rather than simply receiving output classifications.

Academically, the project contributes to the growing literature on multidimensional poverty analysis and regional stress modeling, offering a replicable and extendable methodology grounded in established composite indicator theory. Practically, it provides Armenia with an innovative analytical tool capable of informing national development programs, social protection strategies, and healthcare resource allocation.

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